OutreachIQ

An AI sales team that finds your customers and writes their first email.

Project explainer — what it is, how it works, and where Microsoft Azure fits in. Prepared as background for the presentation.

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1 · What is OutreachIQ?

OutreachIQ is a web app that turns a single company website into a list of real potential customers, each with a personalized outreach email ready to send.

You paste your company's URL. The system reads your site to understand what you sell and who you sell to, then goes out and finds matching companies on the web, researches each one, and drafts a tailored cold email grounded in your own product and case studies.

The core idea: a team of specialized AI "agents", each doing one job, coordinated into a pipeline — and smartly split across two AI providers so quality stays high while cost stays low.

2 · The problem it solves

- Sales reps spend 20–30 minutes per lead researching a company and writing one email.
- Most outreach ends up generic, so it gets ignored.
- Finding which companies are even worth contacting is slow and manual.

OutreachIQ automates that whole top-of-funnel: discover → research → qualify → write.

3 · How it works (end to end)

Setup (once): You paste your company URL. A *self-analyzer* scrapes the site, extracts what you sell + value props, and infers your Ideal Customer Profile (ICP). All of this is loaded into a **knowledge base** that grounds every later step.

Discovery: The system finds **real prospect companies live from the web**. From your ICP it generates search queries, searches the web (via Apify Google Search), extracts the actual companies from the results, and scores each against your ICP. It searches **locally** (your company's country) first, then expands **internationally**.

Per-prospect pipeline: For each top prospect, five agents run in sequence:

Agent	What it does	Runs on
Researcher	Pulls company facts and buying signals	Groq (cheap)
Fit Scorer	Scores the company against your ICP (0–100)	Azure (grounded)
Pain Mapper	Maps their likely problems to your solution + cites a case study	Azure (grounded)
Hook Finder	Picks the best personalization angle	Groq (cheap)
Email Drafter	Writes the outreach email in your voice, with citations	Azure (grounded)

Live view: The user watches the agents work in real time (streaming), with a live cost meter, then reviews the finished emails. Nothing is ever auto-sent — emails are drafts.

Where Microsoft Azure is used

Azure AI Foundry is the brain of the project.

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4 · Azure is the core AI + knowledge layer

Everything that requires real judgment or must be grounded in your company's facts runs on **Microsoft Azure AI Foundry**. The project is built around a Foundry project hosted on an Azure for Students subscription.

Azure services & components we use

Azure component	What it is	How OutreachIQ uses it
Azure AI Foundry (project)	Microsoft's platform for building AI apps + agents	Hosts all our models and the knowledge base; single project/resource for the app
gpt-4.1-mini (model deployment)	A deployed Azure OpenAI chat / reasoning model	Powers the judgment work: ICP inference, fit scoring, pain mapping, email writing
text-embedding-3-small	An Azure OpenAI embedding model	Turns the company docs into vectors so they can be searched by meaning
Knowledge base (vector store)	A Foundry-hosted vector index with file search	Stores your product, ICP and case-study docs; grounds answers and returns citations
Responses API + file_search	Azure's API for grounded, tool-using model calls	Every grounded step retrieves from the knowledge base and cites its sources

What the knowledge base does (the graded part)

When the self-analyzer reads your website, the content is chunked, embedded with **text-embedding-3-small**, and stored in a **Foundry vector store**. Later, when an agent needs to score a prospect or write an email, it calls the **Azure Responses API** with the **file_search** tool — the model pulls the most relevant chunks from the knowledge base and answers using them, returning **citations** (e.g. which doc it used). This is why the emails reference your real case studies instead of making things up.

Where Azure lives in the code

- **backend/config.py** — builds the Azure Foundry client (key auth, OpenAI-compatible v1 endpoint).
- **backend/kb_build.py** — creates the vector store, uploads docs, runs grounded file_search queries.

- **backend/analyzer.py** — uses the Azure model to extract the company profile + ICP from the scraped site.
- **backend/router.py** — *ask_smart()* calls the Azure chat model; *ask_kb()* calls the Azure Responses API + knowledge base.
- **backend/agents.py** — fit scorer, pain mapper and email drafter all run through Azure.

Multi-model orchestration (why this matters)

OutreachIQ deliberately routes work across two providers:

Provider	Role	Cost
Microsoft Azure	Judgment + grounded writing (fit, pain, email, ICP, KB)	Paid (student credit)
Groq (open model)	High-volume bulk work (research, extraction, scoring, discovery)	Free tier

The dashboard shows the live cost split: Azure carries essentially all of the paid cost while Groq handles the bulk for free — the same quality for a fraction of the spend. That routing is the project's core engineering argument.

5 · The rest of the stack

Layer	Technology	Purpose
Frontend	React + Vite + Tailwind	The UI: setup, live run, prospects, dashboard
Backend	Python + FastAPI	Runs the agent pipeline, streams progress (SSE)
Database	SQLite	Stores runs, prospects, emails, per-agent cost trace
Cheap model	Groq (gpt-oss-20b)	Bulk research / extraction / scoring
Prospect discovery	Apify (Google Search)	Finds real prospect companies live on the web

6 · What you see in the demo

- Paste a real company URL → the site is analyzed and the knowledge base is rebuilt live.
- Start a run → agents stream in real time with a live Azure-vs-Groq cost meter.
- Review each prospect: research, fit score, pain points, and a grounded email with citations.
- Dashboard: prospects found, average fit, cost split, and run history.

One-line summary: OutreachIQ turns a company URL into qualified prospects and ready-to-send, grounded outreach emails — built on Microsoft Azure AI Foundry (models + knowledge base) with cost-smart multi-model routing.